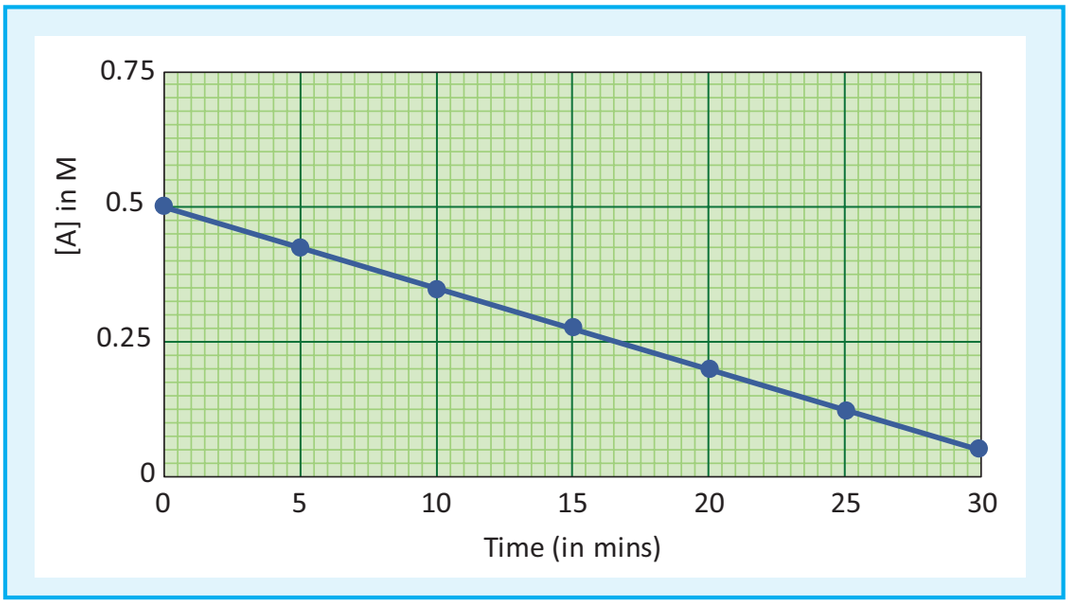


Fig 7.4 : A plot of [A] Vs time for a zero order reaction $A \longrightarrow$ product with initial concentration of [A] = 0.5M and $k = 1.5 \times 10^{-2} mol^{-1} L^{-1} min^{-1}$

Examples for a zero order reaction:

1. Photochemical reaction between H_2 and Cl_2
   $H_2(g) + Cl_2(g) \xrightarrow{h\nu} 2HCl(g)$
2. Decomposition of N_2O on hot platinum surface
   $$N_2O(g) \rightleftharpoons N_2(g) + \frac{1}{2}O_2(g)$$
3. Iodination of acetone in acid medium is zero order with respect to iodine.
   $$CH_3COCH_3 + I_2 \xrightarrow{H^+} ICH_2COCH_3 + HI$$
   $$\text{Rate} = k\,[CH_3COCH_3]\,[H^+]$$

General rate equation for a nth order reaction involving one reactant [A].

$A \longrightarrow$ product

Rate law $\frac{-d[A]}{dt} = k[A]^n$

Consider the case in which $n \neq 1$, integration of above equation between $[A_0]$ and [A] at time $t = 0$ and $t = t$ respectively gives $\frac{1}{[A]^{n-1}} - \frac{1}{[A_0]^{n-1}} = (n-1)kt$

7.6 Half life period of a reaction:

The half life of a reaction is defined as the time required for the reactant concentration to reach one half its initial value. For a first order reaction, the half life is a constant i.e., it does not depend on the initial concentration.

The rate constant for a first order reaction is given by

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

at $t = t_{1/2}$; $[A] = [A_0]/2$

$$k = \frac{2.303}{t_{1/2}} \log \frac{[A_0]}{[A_0]/2}$$

$$k = \frac{2.303}{t_{1/2}} \log 2$$

$$k = \frac{2.303 \times 0.3010}{t_{1/2}} = \frac{0.6932}{t_{1/2}}$$

$$t_{1/2} = \frac{0.6932}{k}$$

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